

Task-Directed Releases Beside a Fixed Prediction Service:

Measuring Additional Sensitive Recovery for Single and Colluding Recipients

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Abstract—An organisation already publishes a prediction service whose outputs downstream recipients consume and cannot be asked to re-accept. We study what it can release *beside* those fixed outputs for one recipient: a channel that must serve a permitted task while limiting what that recipient, and that recipient colluding with another, can newly infer about sex and race. We formalise this purpose-specific release problem, give it an implemented interface and measurement contract—per-recipient and coalition views, independently fitted attackers, additional disclosure always reported against the fixed service baseline, and negative increments retained unclipped—and use it to evaluate a task-directed finite release optimised under explicit conditional-disclosure constraints. On repeatedly used development data, the selected release improves permitted-task loss over the strongest channel this project had by 0.0163 nats with paired household bounds excluding zero, and all eight sensitive point estimates favour it; but seven of the eight adjusted non-inferiority bounds do not resolve, so the registered competitiveness conjunction is not established, and a matched deterministic release without the stochastic machinery performs comparably. We then evaluate two frozen operating points prospectively on a previously unused survey year under a lock created before any final label was read. Both pass all eight registered sensitive-recovery clauses against the prior channel and both fail both task clauses: their task upper bounds are below zero but above the registered 0.003-nat minimum, so each passes eight of ten clauses and neither establishes the full claim. Against matched randomisation and withholding controls the randomised release buys task loss at the price of significantly higher measured recovery on six of eight endpoints, so no frontier advantage for randomisation is established. We report what the comparison does not isolate: the task objective used authorised supervision the comparator never received, and neither randomisation nor coalition conditioning earned its registered attribution. We also correct published claims of our own, including a linear-erasure guarantee that a counterexample refutes, and state precisely which guarantees are finite-model rather than population statements.

I. INTRODUCTION

A bank collects mortgage applications and uses them for several purposes. An underwriter must not see applicant race; a fair-lending auditor must. A pricing analyst sits somewhere in between. Each purpose admits a different set of permissible attributes, and this is the dilemma of *purpose limitation* codified in data-protection regulation and, at the representation level, of contextual integrity.

Most of the machine-learning literature attacks this problem by owning the representation: it edits, projects or retrain

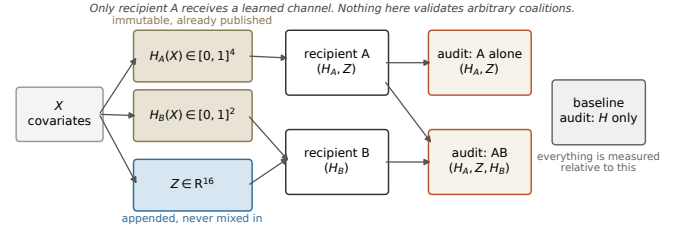


Fig. 1. The release interface. H_A and H_B are already published and are never modified. A channel is appended to recipient A's wire alone. Three views are audited: A alone, the coalition AB, and the H-only baseline against which every increment is measured.

until an attribute is hard to recover. In deployment the harder version is often the one where *the organisation has already published something*. A prediction service is live, its outputs are inside the recipients' systems, and they cannot be recalled or renegotiated. The only decision left is what to place *beside* the fixed outputs, for whom, and how to measure what that addition gives away.

This paper formalises and evaluates that release problem. A service publishes probability vectors H_A to one recipient and H_B to another; a mechanism may add a channel for the first recipient only. The permitted task is a residence-mobility prediction that recipient is authorised to perform; the protected targets are sex and race, forbidden to both recipients individually and to the two acting together.

Contributions:

- **The release problem and its measurement contract (§II):** per-recipient and coalition audit roles, independently fitted attackers, additional disclosure always reported against the fixed service baseline, negative increments retained unclipped, per-endpoint results never averaged.
- **An audit caution that survives checking (§III):** the pooled one-hot regression score is a prevalence-weighted average of per-class scores, so it can conceal a much larger rare-class direction that a per-class (dominant-axis) audit exposes; we state the precise limits of the linear guarantees it is often paired with, including a tight bound for approximate composition.
- **A task-directed finite release (§V):** an instance of

an established convex leakage–distortion programme for one recipient, with its optimisation assumptions, the coarse conditioning it actually uses, and an exact fixture separating what it can and cannot do.

- **A controlled evaluation** (§VI) in which the selected release improves the permitted task over the strongest prior channel with bounds excluding zero, does *not* establish the registered competitiveness conjunction, and is matched by a simpler deterministic baseline—together with the supervision asymmetry that comparison does not isolate.
- **A prospective test on a previously unused survey year** (§VII): two frozen operating points, a lock created before any final label was read, and a ten-clause registered decision that both releases fail at eight of ten—passing every sensitive clause and neither task clause.
- **Corrections to our own published claims** (§C), including a universal regression-to-classification guarantee that an exact counterexample refutes.

II. THE RELEASE PROBLEM AND THE MEASUREMENT CONTRACT

a) Views: Each person has covariates X , sensitive attributes S (sex, 2 classes; race, 9 classes) and task labels. Two services are frozen and published first: $H_A(X) \in [0, 1]^4$ and $H_B(X) \in [0, 1]^2$. A mechanism may add a channel Z for A only, so A holds (H_A, Z) , B holds H_B , and the coalition holds (H_A, H_B, Z) . The policy authorises A for the residence task and forbids sex and race to A , to B and to the coalition.

b) Preservation is architectural; accuracy is not: The channel is concatenated to H_A and never mixed into it, so preservation of the published service is an identity rather than a bounded perturbation, and we check it bitwise on every released view: in the prospective evaluation of §VII the service bytes were identical across all 30 family/anchor interfaces. **That is a statement about bytes, not about accuracy.** The same frozen service, scored on a different survey year, has its own error: on the 2016 final pool its mean accuracy was 0.874 for income, 0.891 for civilian work and 0.760 for public coverage. A recipient who was promised unchanged outputs gets exactly that; a recipient who assumes unchanged *performance* across years does not, and the two claims are never merged here.

c) Threat model: The adversary holds labelled data from the release year, fits attack families on its own pool, selects on a validation pool, and sees only its legal view. It knows the mechanism, the codebook and the channel law; what it does not hold is the person-specific draw. Attacks are fitted on the *unrounded continuous* service predictions, never on a coarsened view, because any conditioning partition used inside an optimiser is a property of that optimiser and not of the released interface.

d) Single-token semantics: The released object is one sampled token per person, drawn once and reused. Publishing the channel row itself, or redrawing per query, is a different mechanism with different disclosure, and we never score one as if it were the other. Expected scoring averages *losses* over

the token distribution, not predicted probabilities; the two differ, and only the first is the expectation of what a recipient experiences.

e) What a number means here: Every recovery figure is the log-loss reduction an independently fitted, validation-selected attacker achieves relative to the same family restricted to H . Absolute recovery—the entropy baseline minus the held-out loss of an attacker fixed before evaluation—is, up to sampling error, a lower bound on the view’s mutual information with the attribute. The increment over the H -restricted fit is a difference of two such lower bounds and bounds $I(S; Z | H)$ in neither direction [1]: a restricted predictor can improve after receiving a deterministic feature $Z = f(H)$ even where $I(S; Z | H) = 0$, and a failed attack bounds nothing from above.

III. WHAT LINEAR AUDITS DO AND DO NOT CERTIFY

The evaluation contract above is a response to a concrete measurement failure we documented in this line of work, and the failure is worth stating because it motivates every audit choice that follows.

a) Aggregate scores hide rare-class leakage: A common compliance reading is an aggregate one-hot regression score $R^2(h, A)$ against a multi-class protected attribute. For an imbalanced attribute this aggregate is a prevalence-weighted convex combination of the per-class one-versus-rest scores,

$$R_{\text{onehot}}^2 = \sum_k w_k R_{\text{OVR},k}^2, \quad w_k = \frac{\pi_k(1-\pi_k)}{\sum_j \pi_j(1-\pi_j)},$$

so a rare class contributes almost nothing to the number a compliance report prints. This is the familiar variance-weighted multi-output R^2 , not a new identity; it holds exactly when both scores share rows, centring and ridge penalty. Auditing the *dominant axis* instead—the largest per-class one-versus-rest score, itself only a lower bound on the best linear sensitive direction—exposes what the aggregate smooths away. On the imbalanced mortgage attribute the gap is large (Table I): mean $R_{\text{DA}}^2 = 0.0609$ against $R_{\text{onehot}}^2 = 0.0192$ over nine multi-class pair-seeds, and in one configuration the aggregate reads 0.027 where the dominant axis reads 0.288, on the rarest class (prevalence 0.9%), for which the aggregate alone cannot bound the class score. Replaying the stored per-class scores reproduces the stored aggregate within 2×10^{-5} in 31 of 33 cells; the two exceptions trace to a single-precision aggregate in the historical evaluation code.

b) Three limits we now state explicitly: First, **a zero linear score does not bound classification.** We previously asserted that zero affine least-squares R^2 forces every linear threshold classifier to majority accuracy. That is false, and a twenty-row counterexample settles it: both conditional feature means are zero, so $\text{Cov}(h, A) = 0$ and the optimal affine predictor is constant with $R^2 = 0$, yet thresholding at $h > 0$ classifies 18/20 correctly against a 50% majority baseline. The claim is withdrawn and the corresponding API refuses to run in the evaluated code. The valid statement concerns losses convex in an affine prediction: zero cross-covariance means no affine predictor improves on the best constant under squared, logistic

TABLE I

AGGREGATE ONE-HOT VERSUS DOMINANT-AXIS AUDITING ON THE MULTI-CLASS PROTECTED ATTRIBUTE, MEAN OVER PAIR-SEEDS OF THE CANONICAL CHECKPOINTS (IN-SAMPLE RIDGE SCORES ON HELD-OUT TEST REPRESENTATIONS). THE AGGREGATE UNDERSTATES THE RECOVERABLE DIRECTION, MOST ON THE IMBALANCED ATTRIBUTE. HISTORICAL EVIDENCE FROM THE ENCODER LINE DESCRIBED IN §IX.

dataset	pairs	mean R^2_{onehot}	mean R^2_{DA}	gap
Adult	12	0.0080	0.0114	+0.0034
HMDA	9	0.0192	0.0609	+0.0417
Diabetes	12	0.0125	0.0179	+0.0054

or any other such loss on that distribution [2]; thresholded accuracy is not such a loss.

Second, **composition is exact only at exact zero**. Two views with exactly zero cross-covariance with the same signal retain exactly zero under concatenation, yet can determine it nonlinearly: with independent signs V, S , the views SV and V each have zero covariance with S and their product is S . Approximate leakage composes with a conditioning penalty: for views with nonsingular covariances, $R^2(h_1, \dots, h_k) \leq \sum_p R^2(h_p) / \lambda_{\min}(R)$, where R is the block-whitened cross-view correlation (a Rayleigh–Ritz bound). It is tight: $h_1 = N + \delta S$ and $h_2 = N - \delta S$ each carry $R^2 = \delta^2 / (1 + \delta^2)$, $\lambda_{\min}(R) = 2\delta^2 / (1 + \delta^2)$, and their difference recovers S .

Third, **more retained rank is not more task performance, and fewer flagged attacks is not low absolute disclosure**. We report absolute recovery beside every increment for this reason.

IV. RELATED WORK

a) Erasure the releaser owns: Linear concept erasure gives an exact guarantee against linear adversaries on a representation the releaser may edit [2], and task-relevant variants preserve authorised covariance [3]. Adversarial fair and transferable representations are due to Madras et al. [4]; adversarial removal is known to leave recoverable information [5], and least-privilege learning has fundamental limits [6]. The closed-form line of Sadeghi et al. [7]–[9] and U-FaTE [10], which conditions its dependence measure on the target label, solve the same shape of problem when the representation is the releaser’s to change. Flexibly fair representations let each task drop a different subset of sensitive attributes from one shared encoder [11]. All of these assume what our setting denies: here the published output is immutable and the only decision is what to append.

b) Finite-alphabet privacy–utility optimisation: The convex programme of §V is established: finite-alphabet leakage–distortion design is convex for a fixed joint law [12], including after quantising to a finite code [13]. Rassouli and Gündüz [14] characterise when perfect privacy is feasible through a nullspace condition and reduce the zero-leakage problem with mutual-information utility to a linear programme; our zero-budget case is conditional and is linear by the same argument. The privacy funnel of Makhdoumi et al. [15] is nonconvex precisely because of a utility constraint we do not impose. A new

release has been designed beside fixed earlier releases under an incremental-leakage budget [16], and per-party and collusion mutual-information constraints on sequential releases were studied recently [17]; since C is fixed, $I(S; Z \mid C) = I(S; Z, C) - I(S; C)$, so our conditional budgets are joint-view budgets with a shifted constant. Side information at the recipient appears in [18], [19], and randomised finite pre-processing under fairness constraints in [20]. We claim no new optimisation result. What is ours is the setting—the conditioning view is a third party’s published prediction service that a coalition partner also holds—the measurement contract, and an evaluated operating point. Related conditional-independence and information-bound machinery we rely on for scoping comes from the kernel conditional-independence literature [21], [22], cross-fitted nuisance estimation [23], and the formal limits of mutual-information estimation [1]. The data are Folktables-style ACS microdata [24].

V. A TASK-DIRECTED FINITE RELEASE

a) Construction: Two residence-supervised teachers, $p(X_A, H_A)$ and $b(H_A)$, define a residual logit $r = \text{logit } p - \text{logit } b$; its 32 quantile cells form a code $T = g(X_A, H_A)$, into which no protected label enters at runtime. T is mapped to a finite output alphabet by a row-stochastic channel Q , and the recipient receives one sampled token. Writing $C_A = H_A$ and $C_{AB} = (H_A, H_B)$ for the conditioning views, the programme is

$$Q \geq 0, \sum_{t,z} p(t) Q_{tz} D_{tz} \quad \text{s.t.} \quad I_Q(S_r; Z \mid C_r) \leq \delta_r,$$

with a cost table D fixed before optimisation. Under the mechanism’s Markov structure $Z - T - (S, C)$ and a fixed fitted law, each conditional mutual information is affine in Q inside a jointly convex relative entropy, so the feasible set is convex and the objective linear. A *local* policy imposes the $r = A$ constraints only; a *coalition* policy adds $r = AB$. The selected utility release is local, so its coalition view is audited but not constrained. We report solver status and replay primal feasibility; no duality gap was recorded, so optimality of the fitted programme is solver-reported, not certified.

b) Attribution, honestly: This is an *established* convex programme applied to a new release contract, not a new optimisation result. The convex finite-alphabet leakage–distortion problem [12], releases beside fixed earlier releases [16], collusion constraints [17] and the nullspace feasibility criterion [14] are prior work; the privacy funnel is nonconvex precisely because of a utility constraint we do not impose. Optimising the decoder jointly with Q , or maximising a task mutual information, destroys the convexity, and we do neither. What is ours is the release contract, the per-recipient and coalition constraint structure, and the evaluated operating point.

c) What the constraint is, and is not: δ_r binds a conditional mutual information in a *fitted finite model* whose conditioning view is a coarsened partition—two cells of H_A for A , four of (H_A, H_B) for AB . It is not a population guarantee and not a statement about the continuous service vectors: in

TABLE II

THE SELECTED RELEASE Q AGAINST J : PAIRED HOUSEHOLD BOOTSTRAP, SIMULTANEOUS ADJUSTED BOUNDS OVER THE FROZEN FAMILY OF 706. THE TASK IMPROVEMENT RESOLVES; SEVEN OF THE EIGHT SENSITIVE ENDPOINTS DO NOT. DEVELOPMENT DATA.

endpoint	wt.	estimate	lower	upper	resolves
<i>permitted task: residence log loss versus J (negative = better)</i>					
residence	unw.	-0.01625	-0.02728	-0.00522	yes
residence	pers.	-0.01639	-0.02917	-0.00362	yes
<i>additional sensitive recovery versus J (negative = less)</i>					
A/sex	unw.	-0.00961	-0.02159	+0.00236	no
A/race	unw.	-0.00567	-0.01717	+0.00583	no
AB/sex	unw.	-0.01103	-0.02199	-0.00007	yes
AB/race	unw.	-0.00394	-0.01552	+0.00765	no
A/sex	pers.	-0.00246	-0.01628	+0.01135	no
A/race	pers.	-0.00482	-0.01760	+0.00797	no
AB/sex	pers.	-0.00620	-0.01887	+0.00647	no
AB/race	pers.	-0.00463	-0.01694	+0.00768	no

a validated fixture, binning a conditioning view drops a full log 2 of measured leakage to exactly zero. All audited attacks therefore use continuous H , and the budgets are never quoted as measured increments over the prior channel.

d) *Why a replacement can beat an extension:* An exact fixture bounds what any “does the new channel add anything?” test can conclude. With H constant, a prior channel $J = (Y, S)$ carrying both the useful bit and the sensitive bit, and a candidate $T = Y$: appending T to J adds exactly zero task information, while *replacing* J by T preserves the task in full and removes the sensitive bit entirely (log 2 nats). A test that measures the first can return zero while the second is strictly better. This is a design example, not an ACS finding.

VI. EVALUATION

a) *Setup:* California ACS one-year public-use microdata, ages 19–34, household-separated pools for representation fitting, task fitting and validation, attacker fitting and validation, and evaluation. Uncertainty is a paired household bootstrap over 5,473 households, aggregated across three anchors that share people, with simultaneous Bonferroni-adjusted bounds over a frozen family of 706 contrasts ($z = 3.97$). All numbers in this section are **development** evidence on repeatedly used pools and are conditional on the fitted releases.

b) *The selected release improves the permitted task:* The utility-route nominee Q reduces residence log loss relative to the strongest channel this project had, J , by 0.01625 nats unweighted and 0.01639 person-weighted, with adjusted bounds excluding zero in both weightings (Table II, Figure 2).

c) *The competitiveness conjunction is not established:* All eight sensitive point estimates favour Q over J , but only one adjusted non-inferiority bound resolves (AB/sex unweighted, upper -0.00007). The registered conjunction therefore fails. **This is not evidence of excess disclosure:** seven endpoints are unresolved at this precision, and an unresolved interval is not a finding in either direction.

d) *A simpler baseline matches it:* D_{17} uses the same code and the same 17-action dictionary with no privacy constraint and no stochastic optimisation. It attains a slightly *better* task point estimate than Q and *lower* sensitive point estimates in

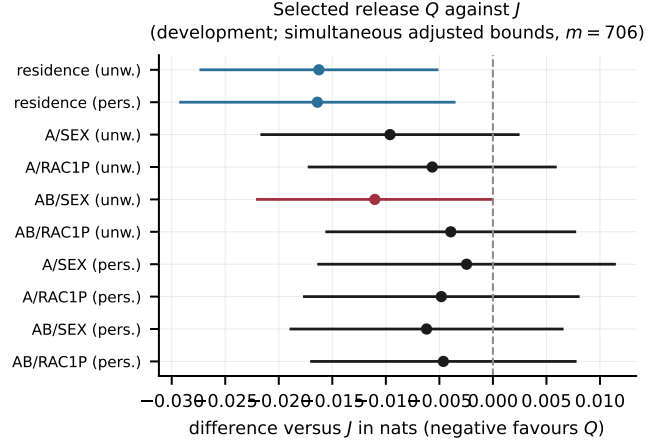


Fig. 2. The same comparison. Every sensitive point estimate favours Q , and seven of eight intervals cross zero: *unresolved*, which is neither demonstrated protection nor demonstrated harm.

TABLE III

RELEASES COMPARED AGAINST J ON THE PERMITTED TASK AND ON ADDITIONAL SENSITIVE RECOVERY (UNWEIGHTED POINT ESTIMATES; POSITIVE TASK VALUES FAVOUR THE RELEASE). H CARRIES NO CHANNEL, SO IT IS WORSE AT THE TASK AND LEAKS LESS. DEVELOPMENT DATA; POINT ESTIMATES ONLY.

release	task vs J	worst sens. vs J
H only (no channel)	-0.02570	-0.00550
J (prior channel)	-0.00000	-0.00000
Q : task-directed, constrained	+0.01625	-0.00246
D_{17} : deterministic, same dictionary	+0.01720	-0.00285
D_{33} : deterministic, larger dictionary	+0.01794	-0.00364
continuous task prediction	+0.01962	+0.00123

seven of eight cells (Table III). All ten corresponding adjusted differences are unresolved, so we do not claim D_{17} is better, and we equally do not claim the two are equivalent. D_{33} , with a larger dictionary, has better task point estimates and mixed sensitive comparisons; its cap feasibility was not established and it is not a dominating method. The honest reading is that on this evidence **the stochastic machinery has not been shown to buy anything over a matched deterministic release.**

e) *Baselines, and what each one was allowed to see:*

The comparison set is deliberately wider than J . *Deterministic action releases* (D_{17} , D_{33}) use the same residence-supervised code and action dictionary with no privacy constraint. A *direct code* release publishes the code index itself. A *continuous task prediction* publishes the real-valued authorised prediction and is the strongest task point in the table, with the worst sensitive comparison of the group. *Supervised erasure adaptations* (LEACE and SPLINCE) were fitted with protected labels on the same standardised inputs plus the clipped frozen residence-teacher logit; SPLINCE additionally preserves residence, income and employment covariance, so it sees authorised task labels that J never saw. Two fitting pools were prepared in advance—one matching the mechanism’s own fresh-label rows and one the broader union—so that no post-outcome

choice between fairness interpretations was needed, with each method’s own complete-case mask disclosed rather than assumed identical. These adaptations completed, but neither supplied an *eligible* nominee for the utility route, which is why the stricter all-family diagnostic is reported as blocked rather than as passed or failed. An unavailable comparator family is not a verdict about that family.

f) What the comparison does not isolate: Q ’s objective used authorised residence supervision. J never received that training objective. The comparison therefore establishes neither unseen-task transfer nor an isolated benefit of the new optimiser, and a supervised task result must not be described as an untouched-task result. The continuous task predictor and the supervised eraser adaptations are reported with their actual label access and pools for the same reason.

g) Neither refinement earned its claim: Neither risk refinement nor coalition conditioning obtained its registered attribution: the nominated release is *local*. Beating J alone cannot attribute a benefit to the coalition machinery.

h) Why an earlier explanation was wrong: A predecessor experiment found that a label-free code of the permitted inputs added nothing measurable once the published predictions were in the recipient’s view, and read this as the published output already carrying the available signal. A subsequent controlled diagnosis separated the two candidate explanations and found the opposite: the *unquantized* input view attains strictly lower residence loss than J , with adjusted bounds below zero, while a 64-state code of the same inputs gives that advantage away. The obstacle was quantization, not redundancy, and the earlier universal reading is withdrawn (§C).

VII. A PROSPECTIVE TEST ON A PREVIOUSLY UNUSED YEAR

Everything above is development evidence on pools this project has reused. We therefore froze two operating points—the randomised release Q and the matched deterministic release D_{17} —and evaluated them on a California ACS 2016 cohort that no earlier stage had touched: 23,684 people in 15,928 households, with the code, release laws, attacker and probe recipes, fitting selections, estimands and inference thresholds locked before any final-partition label was read. No mechanism was fitted or selected on 2016 outcomes. The registered primary claim is a ten-clause conjunction: two task clauses requiring an uncertainty-adjusted improvement of at least 0.003 nats over J , and eight sensitive-recovery clauses capping added recovery at +0.001 nats.

Both releases fail the conjunction, and both fail it in the same way (Table IV, Figure 3). Each passes **all eight** sensitive clauses: every upper bound lies below zero, which is stronger than the registered cap. Each fails **both** task clauses. Q ’s task upper bounds are -0.001505 and -0.001871 ; D_{17} ’s are -0.002913 and -0.002951 ; the registered requirement was ≤ -0.003 . So each release passes eight of ten clauses and neither establishes the complete claim.

The distinction matters and follows from the same bounds. Both task upper bounds are strictly below zero, so the data do

TABLE IV
PROSPECTIVE ACS 2016, REGISTERED PRIMARY CLAUSES. UPPER BOUNDS ARE ONE-SIDED 97.5%, FROM A COMMON 10,000-DRAW PAIRED HOUSEHOLD BOOTSTRAP OVER THE SHARED HOUSEHOLD POPULATION, RE-DERIVED HERE AS ESTIMATE + $z \cdot \text{SE}$. BOTH RELEASES PASS 8/10.

endpoint	wt.	estimate	upper bd.	threshold	clause
Q (randomised)					
residence (task)	unw.	-0.00327	-0.00151	-0.003	fail
residence (task)	pers.	-0.00400	-0.00187	-0.003	fail
A/sex	unw.	-0.01284	-0.01089	+0.001	pass
A/sex	pers.	-0.01322	-0.01080	+0.001	pass
A/race	unw.	-0.03840	-0.03375	+0.001	pass
A/race	pers.	-0.03657	-0.03105	+0.001	pass
AB/sex	unw.	-0.00887	-0.00724	+0.001	pass
AB/sex	pers.	-0.00938	-0.00736	+0.001	pass
AB/race	unw.	-0.02670	-0.02284	+0.001	pass
AB/race	pers.	-0.02870	-0.02400	+0.001	pass
D_{17} (deterministic)					
residence (task)	unw.	-0.00474	-0.00291	-0.003	fail
residence (task)	pers.	-0.00514	-0.00295	-0.003	fail
A/sex	unw.	-0.01243	-0.01046	+0.001	pass
A/sex	pers.	-0.01211	-0.00968	+0.001	pass
A/race	unw.	-0.03786	-0.03319	+0.001	pass
A/race	pers.	-0.03681	-0.03122	+0.001	pass
AB/sex	unw.	-0.00904	-0.00738	+0.001	pass
AB/sex	pers.	-0.00928	-0.00723	+0.001	pass
AB/race	unw.	-0.02462	-0.02065	+0.001	pass
AB/race	pers.	-0.02797	-0.02316	+0.001	pass

support *a* task improvement over J on this fresh year; what they do not support is the *registered minimum* improvement. That is a statement about the size of the effect relative to a margin fixed in advance, not a rescue of the claim by a new criterion, and the margin is not relaxed here. The margin itself was not unreachable: a precision projection made before scoring, from archived development losses, put the attainable standard error at 0.0009–0.0016 (the realised value was 0.0009–0.0011) and the pass probability near one *if the development effect size carried over*. It did not. The development improvement over J was -0.0163 nats; the prospective estimate is -0.0033 , about a fifth of it. **The conjunction failed because the effect shrank on a fresh year, not because the test was underpowered for the effect it expected**—which is the most informative thing this evaluation produced, and a caution about development effect sizes generally. Equally, the eight passing sensitive clauses are a real result on this attacker slate and these access views—not a universal privacy guarantee, not evidence about arbitrary attackers, and not a coalition-specific property. **Their inferential scope is also narrower than it looks when they are quoted alone.** The primary decision is an intersection–union test: each clause is checked at its own one-sided level, which is what makes the *conjunction* valid without a simultaneous correction. It does not follow that the eight sensitive clauses, lifted out and highlighted on their own, carry simultaneous 95% coverage as a set. We report them as eight per-clause results inside a conjunction that failed, and we do not restate them as a jointly certified family.

Against matched simple controls, no frontier advantage appears. In a separate, multiplicity-corrected family of 70 endpoints, Q has lower task loss than a 75% randomised-response control (-0.011652 , -0.011170) and a 75% withhold-

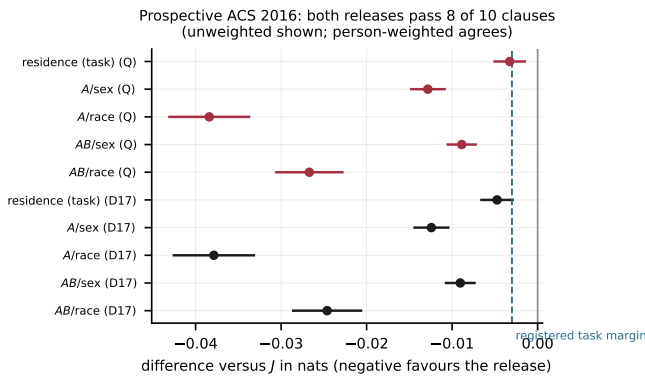


Fig. 3. The same decision. Every sensitive interval lies below zero; both task intervals lie below zero but cross the registered margin. Prospective evaluation on a previously unused year.

ing control (-0.007055 , -0.006875) under both weightings—but it also has *significantly higher* measured sensitive recovery on **six of eight** endpoints against each. A complete task-and-privacy benefit over these controls is therefore not established, and task performance alone does not imply dominance. Against the deterministic D_{17} and D_{33} , Q is *significantly worse* on unweighted task loss (intervals excluding zero) while the person-weighted contrasts are *unresolved*, and no sensitive endpoint shows Q superior to either control—all eight are unresolved in both directions. So **randomisation is not shown to beat the matched deterministic control** on this evidence, and it is not shown to be worse on disclosure either. Direct supervised and erasure-adaptation controls achieve better task loss than Q at different measured disclosure costs.

What this evaluation is not. The frozen release family was built with task and sensitive labels during earlier development, so this is a fresh-year evaluation of a *supervised* fixed release family, not evidence of transfer to an unseen task. It does not replace the 2018 development programme’s own failed decision, and the two are reported separately throughout. The panel completed without candidate replacement or outcome-driven stopping: 156/156 fit units, 156/156 score units, zero unit failures, zero quarantined scores, 10,000/10,000 accepted bootstrap draws. An independent replay reproduced every stored per-person expected loss to 3.6×10^{-15} and all 204 point estimates to 1.4×10^{-17} , and the published service outputs were byte-identical across all 30 interfaces.

VIII. LIMITATIONS, AND A DIRECT ANSWER

Development versus prospective evidence. The 2018 pools have been used repeatedly across this line of work; three anchors and simultaneous intervals do not restore independence from prior research. The 2016 evaluation of §VII is prospective and locked, but it evaluates a release family whose teachers and code were built with labels during that earlier development, so it is a fresh-year test of a supervised fixed family rather than an unseen-task result. The two bodies of evidence are never pooled.

Failing a conjunction is not a null. Both 2016 releases failed both task clauses while their task upper bounds sat below zero; that is a shortfall against a pre-registered margin, not a demonstration that the task effect is zero. Conversely, passing eight sensitive clauses is a measured result for the registered attacker slate, roles and access views, not a guarantee against other attackers.

Unresolved is not equivalence, and not harm. Seven of eight sensitive comparisons are unresolved. We do not convert them into either claim.

Fitted-model budgets are not population guarantees, coarse conditioning is not conditioning on the continuous service vectors, and negative finite-attacker increments do not demonstrate negative conditional mutual information.

Attacks bound from below only. Three searches for a stronger attacker across this project each failed to beat the standard slate, which bounds what those searches found, not what exists.

Audits do not transport across hardware below about 0.003 nats, so comparisons use same-host references.

a) *What was established:* A task-directed release beside a fixed service improved the permitted task over the strongest channel we had, with bounds excluding zero on development data; and on a previously unused year, two frozen operating points each passed all eight registered sensitive-recovery clauses against that channel, with task improvements whose upper bounds were below zero. **What was not established:** the registered conjunction in either year—on 2016 both releases missed the 0.003-nat task margin—any advantage of the stochastic machinery over a matched deterministic release, any complete benefit over matched randomisation or withholding controls, any coalition-specific benefit, and any unseen-task claim.

IX. RELATION TO THE EARLIER ENCODER LINE

This work descends from a purpose-conditioned representation line in which one shared frozen, randomly initialised backbone carried per-purpose adapters, each warm-started from a closed-form erasure projection [2] and trained under a per-purpose linear leakage bound. That setting differs from this one in its release contract: there the releaser owned and re-emitted the representation, here the service output is immutable and the decision is what to append. The auditing result of §III and the composition limits come from that line and survive checking; its training results are not re-used as evidence for the present mechanism, because they tested a different algorithm against a different claim. Appendix A gives the correspondence.

OPEN SCIENCE

Every number in this paper is generated from committed evidence at pinned commits, and each printed digit traces to a (commit, path, SHA-256) triple recorded in the artifact. The artifact contains the table and figure generators, the independent verification scripts, the claim ledger, the correction log, synthetic fixtures, schemas, aggregate result tables and provenance hashes, together with a recipe for obtaining the

public ACS microdata this work uses. No restricted data is redistributed and no individual-level records are included. Anonymised artifacts will be shared within the window the call specifies.

LLM USAGE CONSIDERATIONS

This work was substantially assisted by a large language model, and describing that as editorial help would be inaccurate. A model was used to propose and refine experimental designs, to write and refactor the experiment and analysis code, to draft and revise this manuscript, and to audit the studies’ own reports against their source code and stored outputs. Registrations, margins and stopping rules were committed before outcomes. Model-generated analyses were checked by independently written scripts that recompute each reported quantity from the smallest machine-readable source, and several statements in this paper exist because that recomputation contradicted an earlier write-up—the withdrawn explanation in §VI and the retired guarantee in §III are two. The authors take full responsibility for every claim, number and omission in this work, including those first drafted by a model; nothing was accepted on the model’s assertion alone, and every load-bearing figure is reproducible from the artifact.

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APPENDIX A

LINEAGE AND WHAT EACH STUDY TESTED

The encoder line and the present release line share an evaluation philosophy and a protected-attribute audit, not

an algorithm. Claims from the encoder line used here are limited to the dominant-axis audit, the convex-combination identity, the retired guarantee and the composition boundary. The encoder line’s training headline totals are same-grid post hoc development evidence: the dual floor and warm-up rule were developed after examining an earlier evaluation on the same grid, and neither was validated on a held-out subset of purpose/attribute/seed triples. A later prospective registration does not retrospectively change that scope.

Two further scope facts belong with those totals. *Checkpoint rule*. Counts differ by which checkpoint is scored: the final iterate gives 56/60 cells meeting the strict linear criterion, of which 7 also clear a non-collapse health check, while a per-seed best-checkpoint rule gives 54/60; a later erase-layer arm reaches 60/60 strict but 0/60 clean. Strict and clean counts are reported separately and under one stated rule, never mixed. *Untested components*. Whether the low-rank adapter or the frozen erasure layer is *necessary* was registered as two ablations that were never run, so no necessity or advantage is claimed for either; the one comparison that exists confounds the erasure layer with the constraint and cannot separate them. Neither this paper nor the release mechanism it evaluates shares an implementation with that line: no combined encoder, adapter and stochastic-release pipeline has been built or evaluated, and the two are reported as separate implementations throughout.

APPENDIX B COUNTS AND ACCOUNTING

The task-directed programme’s primary matrix is 81 nominal maps (54 positive, 18 zero, 9 unconstrained); an independent replay covered all 81. Of 162 nominal maps in the full matrix, 126 were scheduled and completed and 36 were prospectively unscheduled under the registered resource schedule; the completed maps reduce to 125 exact kernel groups. There are 297 current audits and evaluations. These are resource and accounting figures, not scientific contributions.

APPENDIX C CORRECTIONS THIS PAPER MAKES

- 1) **The universal regression-to-classification guarantee is withdrawn.** Zero affine R^2 can coexist with 90% threshold accuracy; the valid statement is the narrower squared-loss one (§III).
- 2) **A shared union eraser does not demonstrate retention of conflicting purposes.** In the recorded comparison the shared union destroys both tasks ($R^2 \approx 0$ on each), while separate per-purpose erasure retains both (0.9996, 0.9997); the restricted arm ties that comparator rather than beating it.
- 3) **“The published output already carries the signal”** is withdrawn: a controlled diagnosis attributes the effect to quantization, not redundancy (§VI).
- 4) **A proposed Sadeghi–Boddeti near-optimality certificate is not an implemented result.** It was explicitly skipped for want of the necessary numerical check and is not claimed here.

- 5) **The encoder line’s backbone was not pretrained.** It was described as pretrained on the union of permitted tasks; in the implementation that produced its results it was a seeded random MLP, frozen at initialisation.
- 6) **A rank floor for single-layer edits is not a floor for the trained adapters.** Erasure below $\sum_{i>r} \sigma_i^2 / \text{tr Cov}(A)$ is impossible for one rank- r multiplicative edit of a fixed representation, but the trained adapters edit every layer with nonlinearities in between; a rank-one edit of the first layer can remove a rank-two sensitive cross-covariance entirely.
- 7) **Exact-zero covariance composition** holds; approximate leakage does not inherit it, and neither implies independence or a nonlinear guarantee.